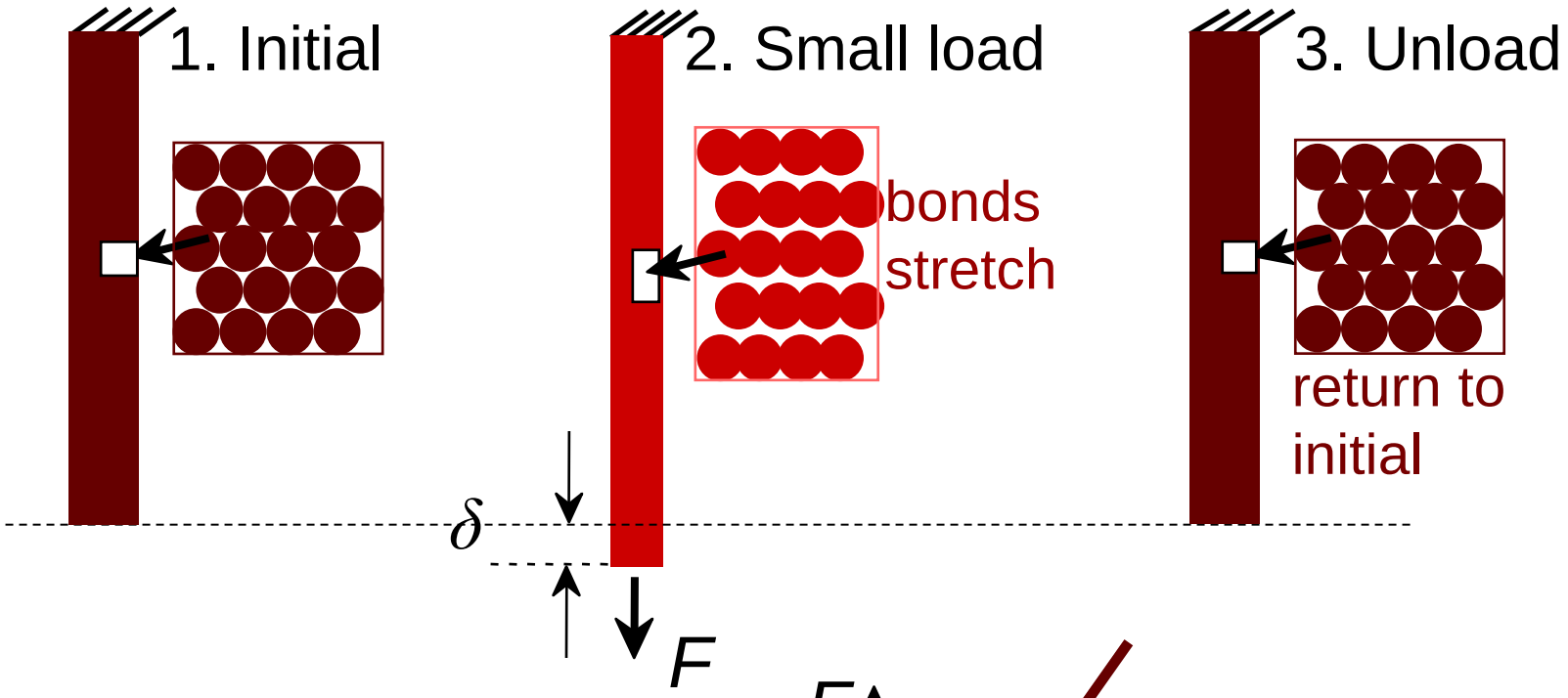
Chapter 8:
Mechanical Properties of Metals

ISSUES TO ADDRESS...

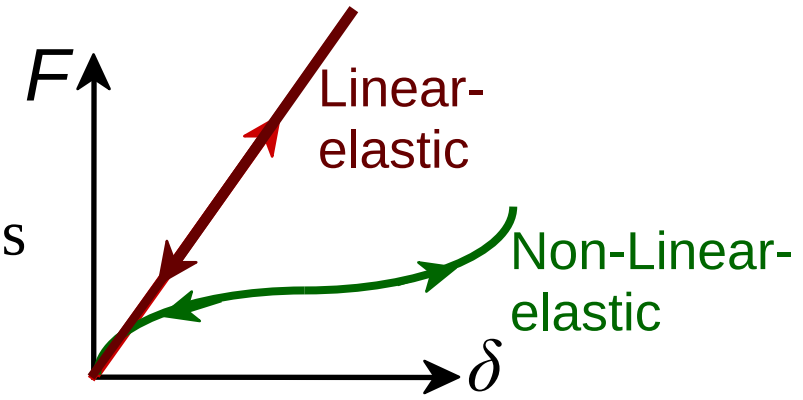
- Stress and strain: What are they and why are they used instead of load and deformation?
- Elastic behavior: When loads are small, how much deformation occurs? What materials deform least?
- Plastic behavior: At what point does permanent deformation occur? What materials are most resistant to permanent deformation?
- Toughness and ductility: What are they and how do we measure them?

Elastic Deformation



Elastic deformation refers to a temporary deformation of a material's shape that is self-reversing after removing the force or load.

Elastic means reversible!



Stress and Strain

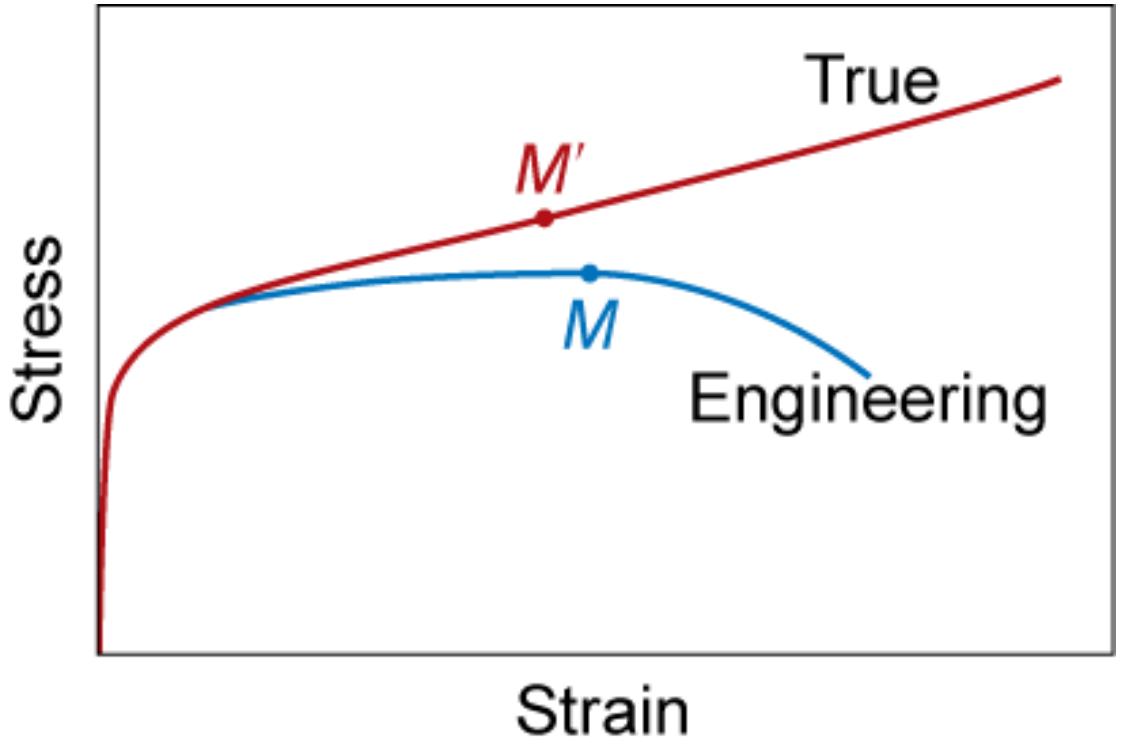
- Engineering stress is the ratio of the applied load to the original cross-sectional area of the material, while engineering strain is the ratio of the change in length to the original length of the material.
- True stress is the ratio of the applied load to the instantaneous cross-sectional area of the material, while true strain is the natural logarithm of the ratio of the change in length to the original length of the material.
- Shear stress, force tending to cause deformation of a material by slippage along a plane or planes parallel to the imposed stress, while Shear strain occurs when a sideways force is exerted on a medium.
- Lateral Strain is defined as the ratio of the change in diameter of a circular bar of a material to its diameter due to deformation in the longitudinal direction.

True Stress & Strain

- True stress $\sigma_T = F/A_i$
- True strain $\epsilon_T = \ln\left(\ell_i/\ell_o\right)$

$$\sigma_T = \sigma\left(1+\epsilon\right)$$

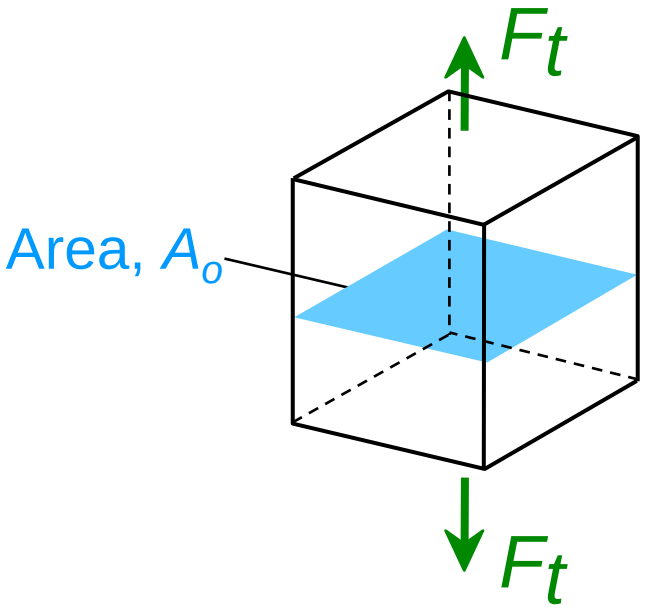
$$\epsilon_T = \ln\left(1+\epsilon\right)$$



Adapted from Fig. 8.16, *Callister & Rethwisch 9e.*

Engineering Stress

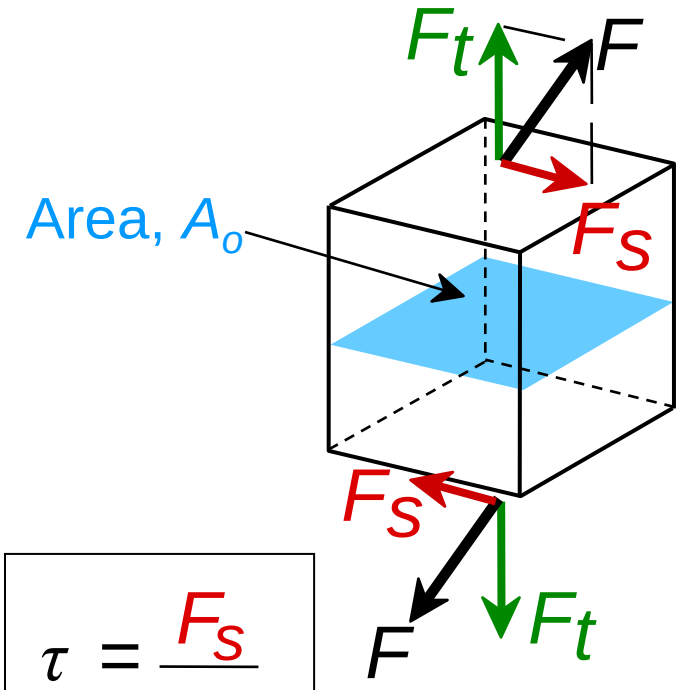
- Tensile stress, σ:



$$\sigma = \frac{F_t}{A_o} = \frac{\text{N}}{\text{m}^2}$$

original cross-sectional area before loading

- Shear stress, τ:



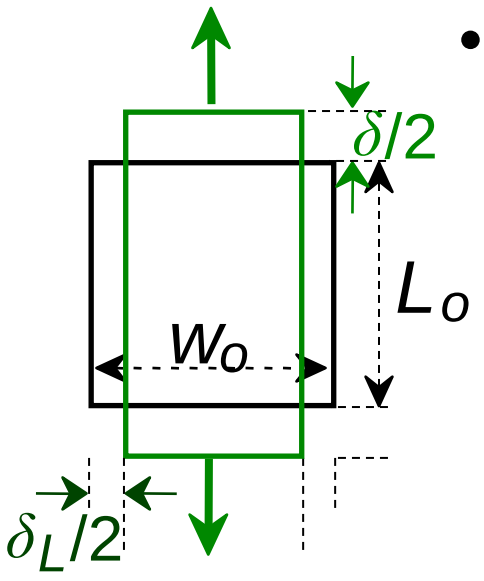
$$\tau = \frac{F_s}{A_o}$$

$\therefore$ Stress has units: N/m^2

Engineering Strain

- **Tensile strain:**

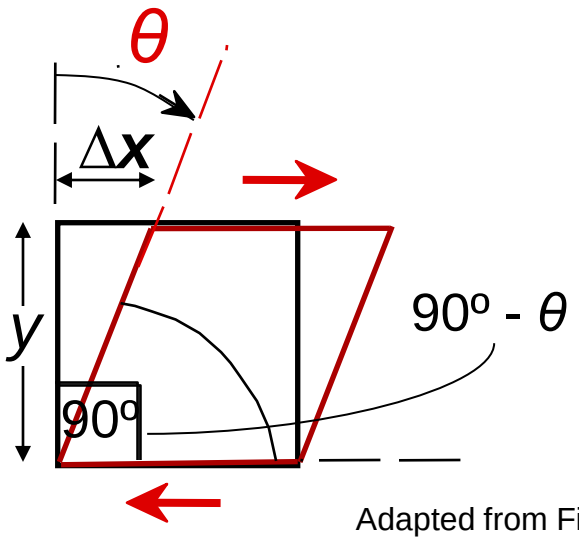
$$e = \frac{\delta}{L_o}$$



- **Lateral strain:**

$$e_L = -\frac{\delta_L}{w_o}$$

- **Shear strain:**



$$\gamma = \Delta x / y = \tan \theta$$

Strain is always dimensionless.

Adapted from Fig. 8.1 (a) and (c), *Callister & Rethwisch 9e.*

Stress-Strain Testing

- **Typical tensile test machine**

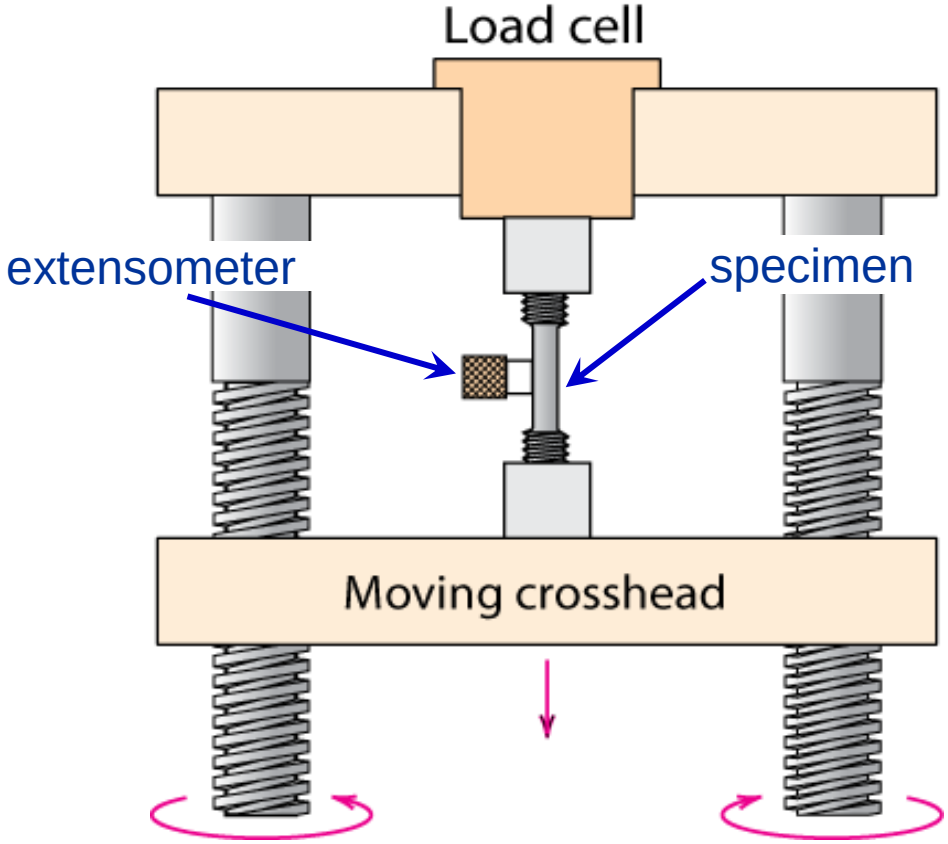


Fig. 8.3, *Callister & Rethwisch 9e.*
(Taken from H.W. Hayden, W.G. Moffatt, and J. Wulff, *The Structure and Properties of Materials*, Vol. III, *Mechanical Behavior*, p. 2, John Wiley and Sons, New York, 1965.)

- **Typical tensile specimen**

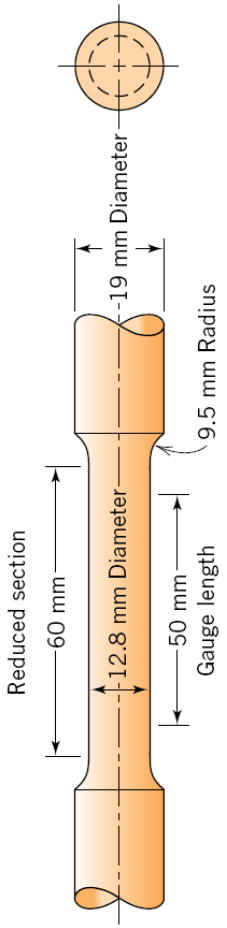
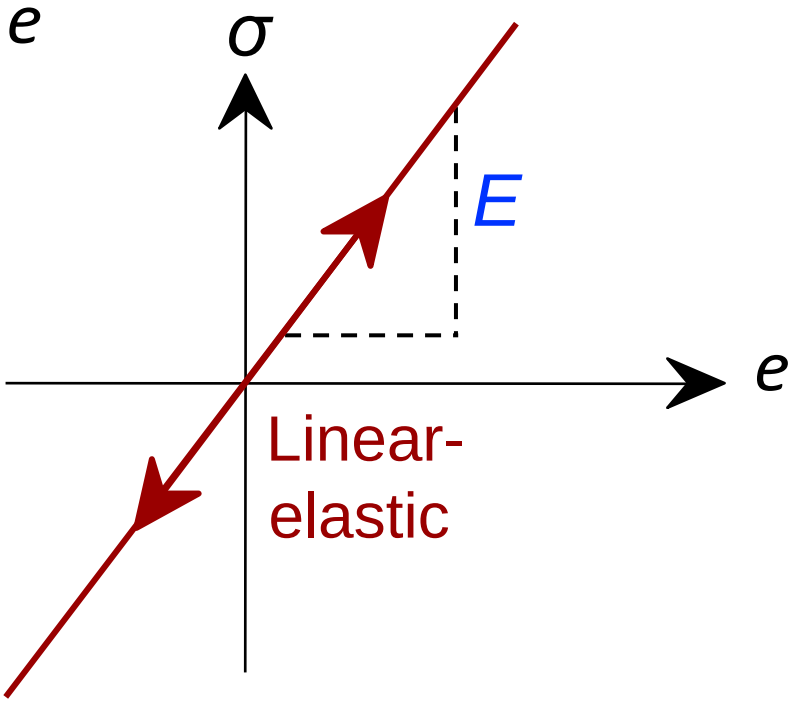


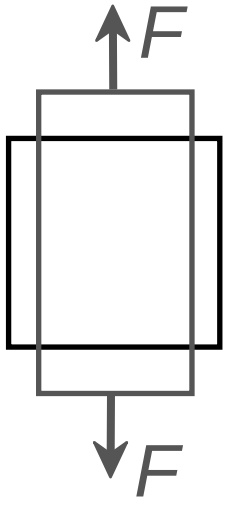
Fig. 8.2, *Callister & Rethwisch 9e.*

Linear Elastic Properties

- **Modulus of Elasticity, *E*:**
  (also known as Young's modulus)
- **Hooke's Law:**

$$\sigma = E\, e$$



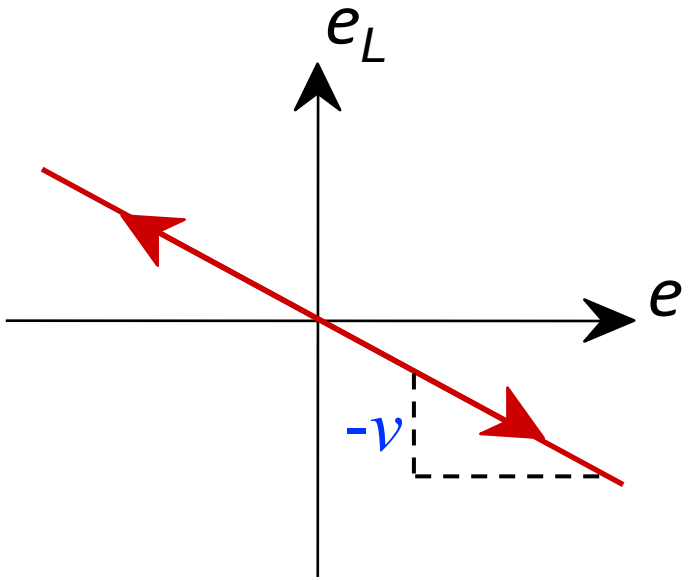


simple tension test

Poisson's ratio, ν

Poisson's ratio, ν: is the ratio of the lateral strain to the tensile strain

$$\nu = - \frac{e_L}{e}$$



metals: $\nu \sim 0.33$

ceramics: $\nu \sim 0.25$

polymers: $\nu \sim 0.40$

Units:

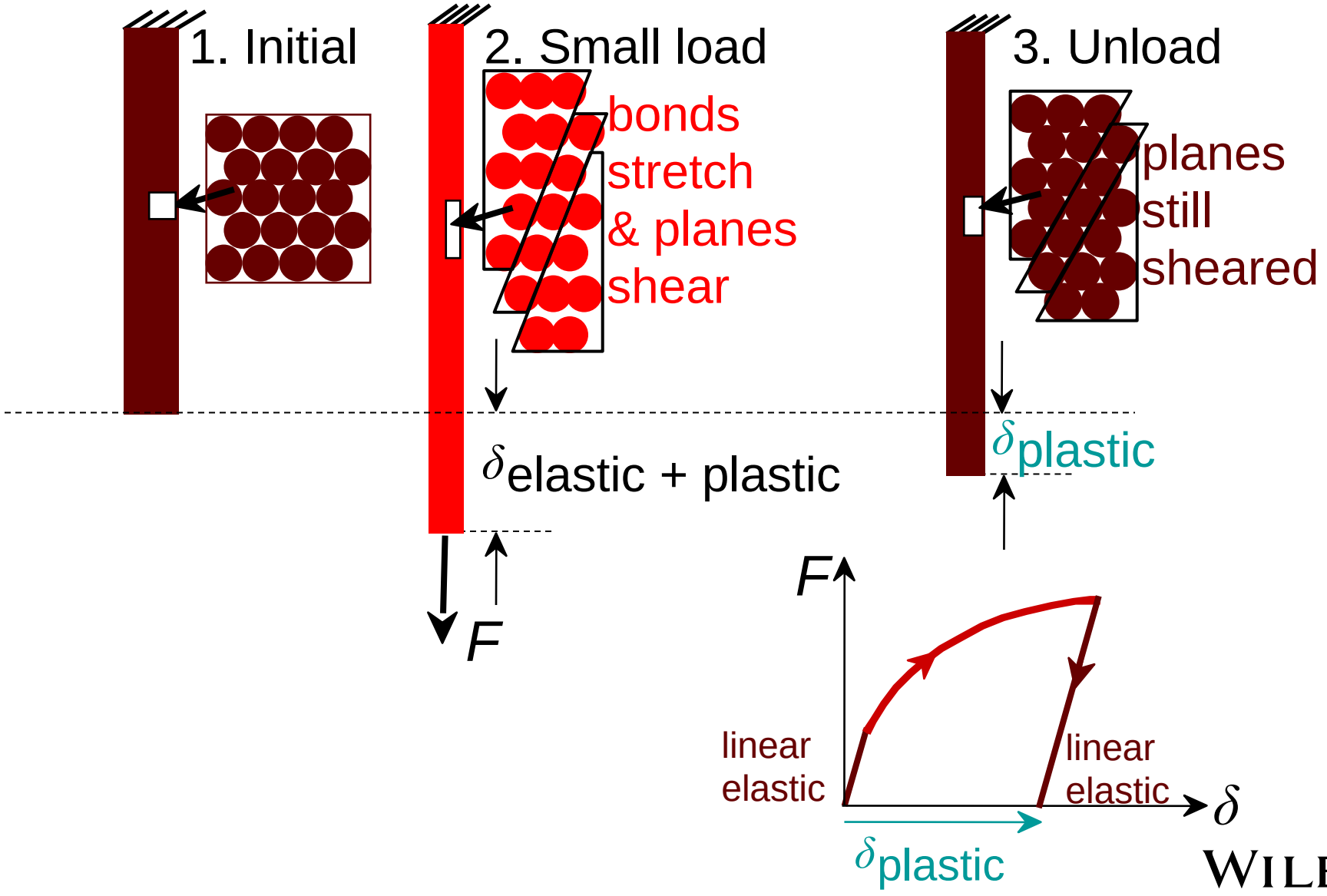
E: [GPa] or [psi]

ν: dimensionless

$\nu > 0.50$ density increases

$\nu < 0.50$ density decreases (voids form)

Plastic Deformation (Metals)

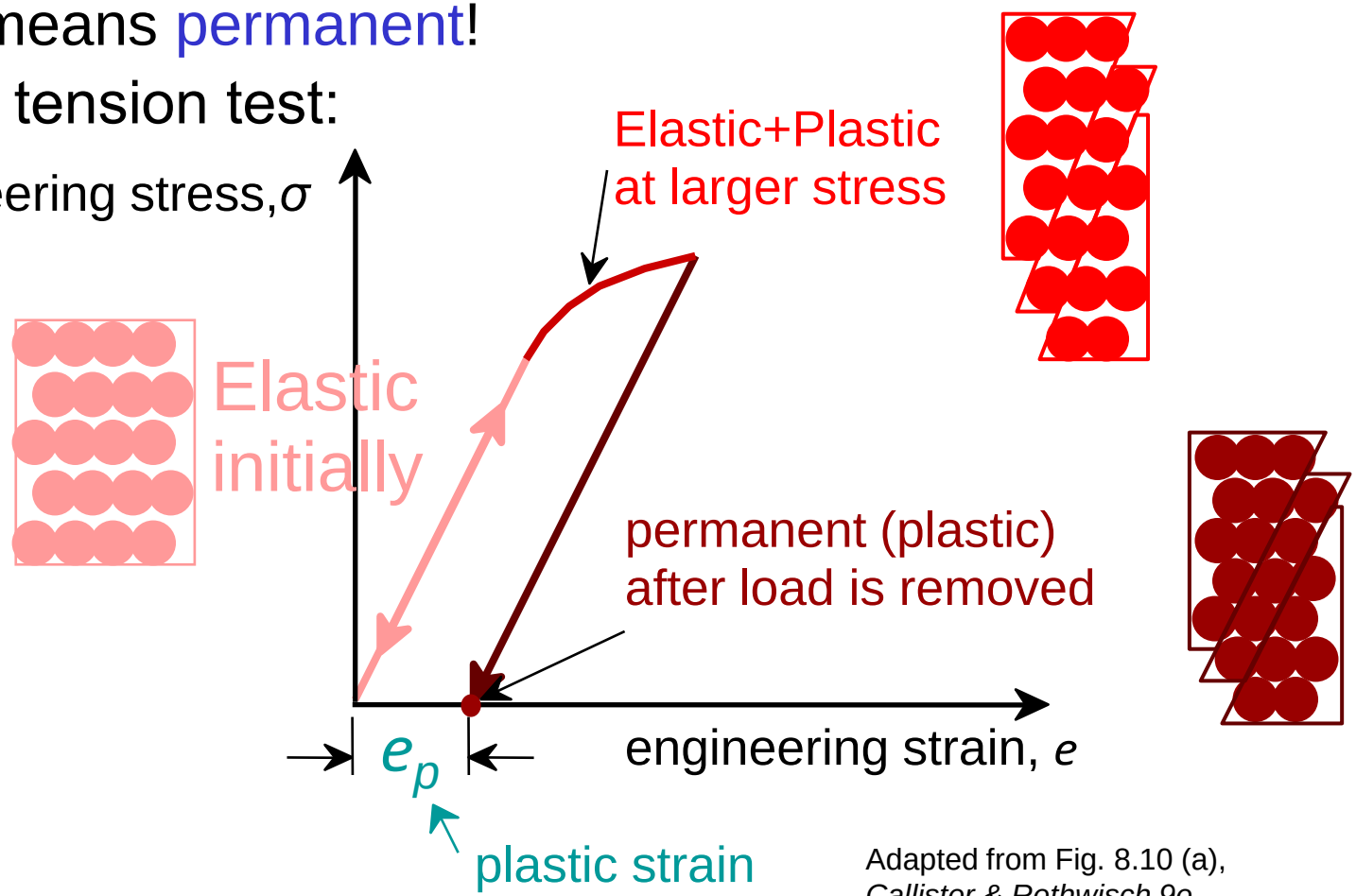


Plastic (Permanent) Deformation

Plastic deformation refers to a permanent deformation of a material's shape that is not reversing after removing the force or load.

Plastic means permanent!

- Simple tension test:

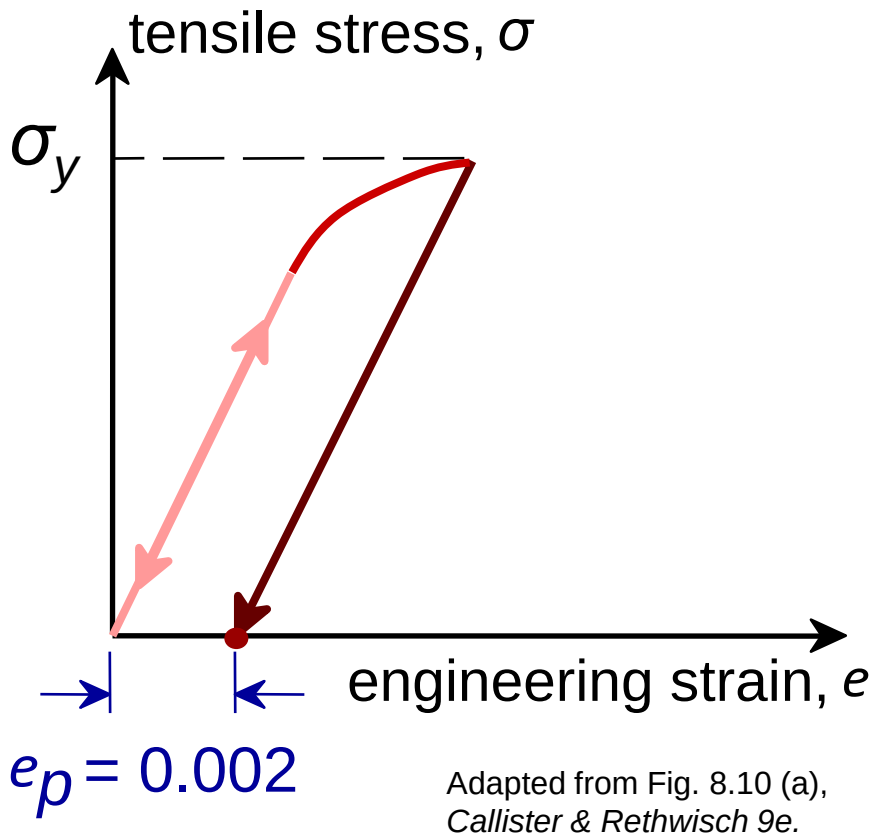


Adapted from Fig. 8.10 (a), *Callister & Rethwisch 9e.*

Yield Strength, σ_y

- **Stress at which *noticeable* plastic deformation has occurred.**

when $e_p = 0.002$



Adapted from Fig. 8.10 (a), *Callister & Rethwisch 9e.*

σ_y = yield strength

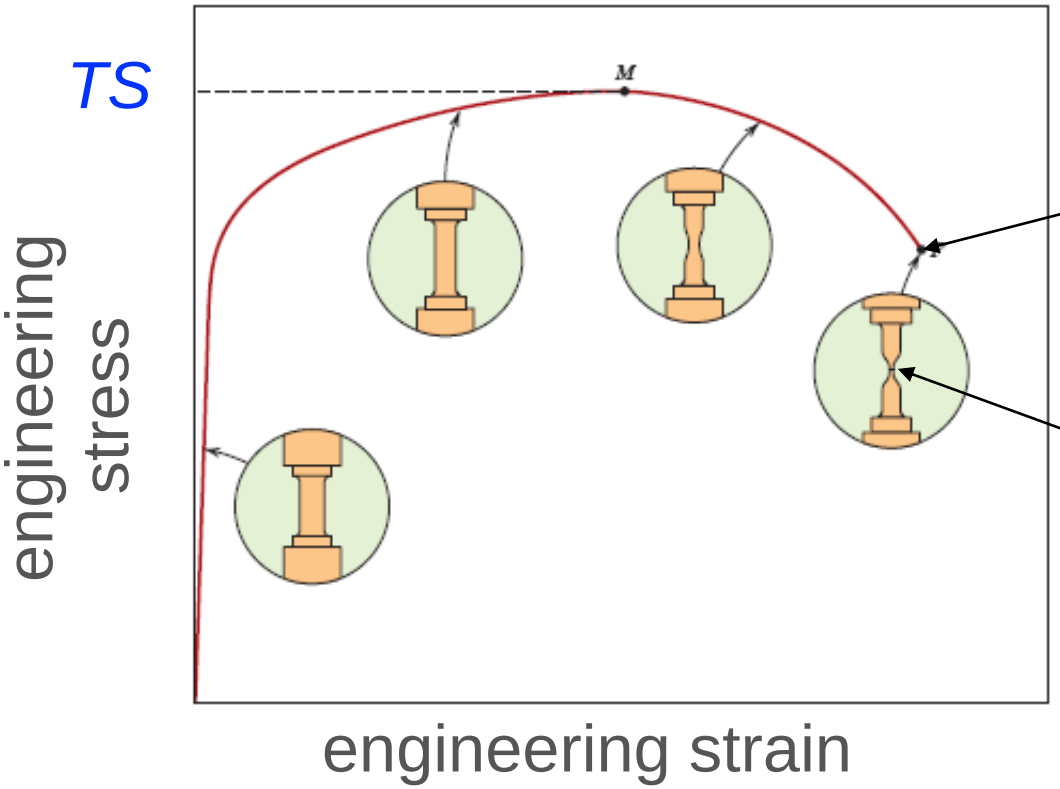
Note: for 2 inch sample

$e = 0.002 = \Delta z/z$

$\therefore \Delta z = 0.004$ in

Tensile Strength, TS

- Maximum stress on engineering stress-strain curve.



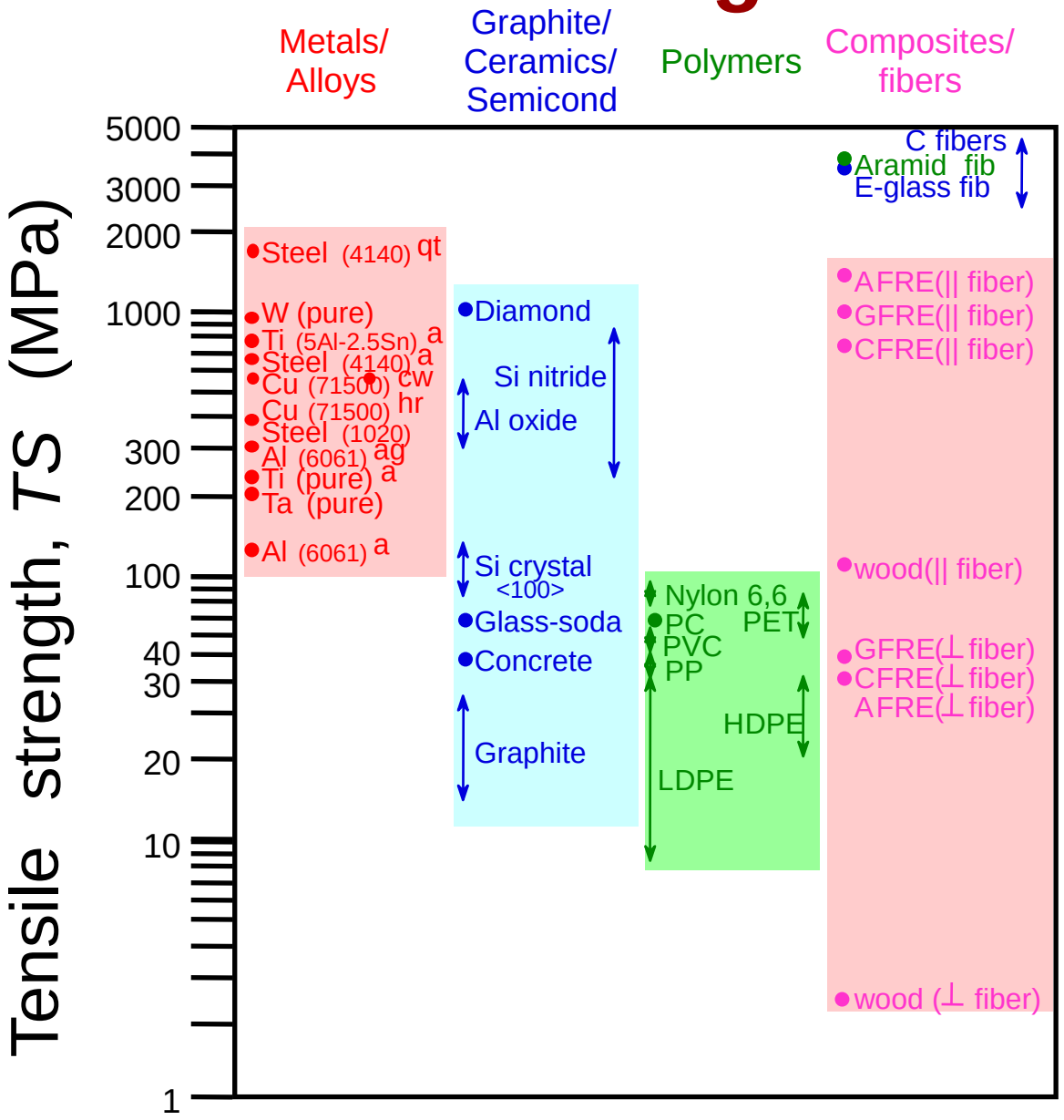
Adapted from Fig. 8.11, *Callister & Rethwisch 9e.*

F = fracture or ultimate strength

Neck – acts as stress concentrator

- Metals: occurs when noticeable necking starts.
- Polymers: occurs when polymer backbone chains are aligned and about to break.

Tensile Strength: Comparison



Room temperature values

Based on data in Table B4, *Callister & Rethwisch 9e.*

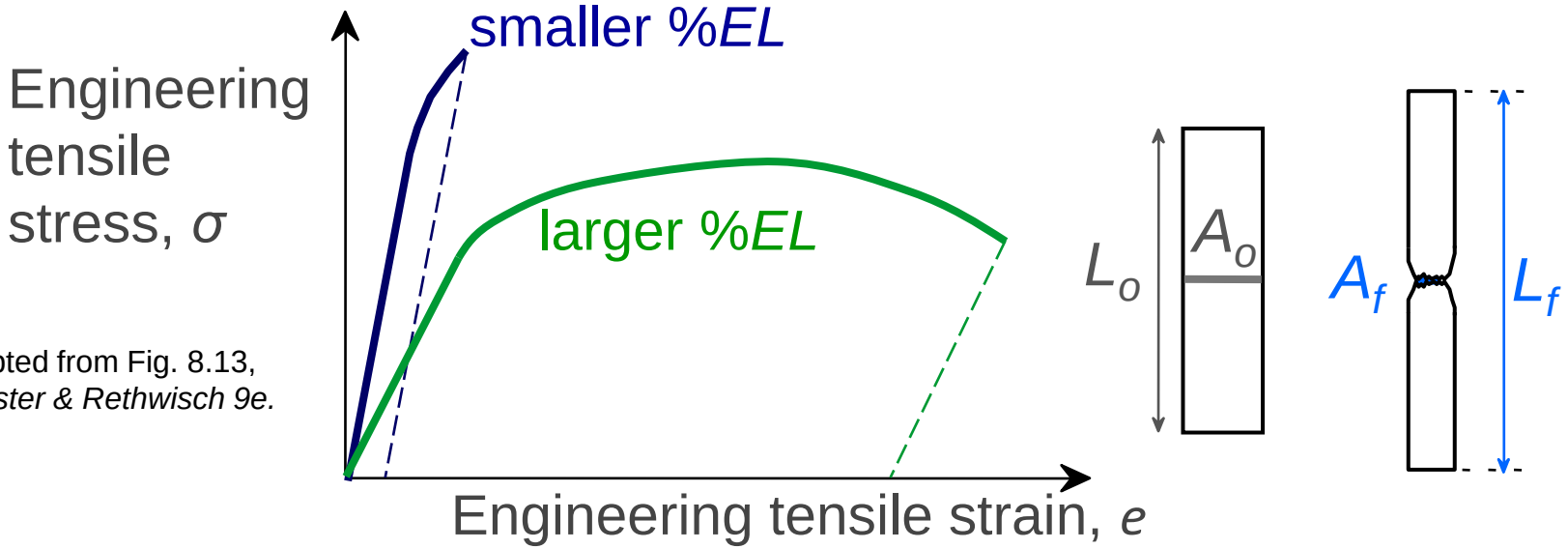
a = annealed
hr = hot rolled
ag = aged
cd = cold drawn
cw = cold worked
qt = quenched & tempered

AFRE, GFRE, & CFRE = aramid, glass, & carbon fiber-reinforced epoxy composites, with 60 vol% fibers.

Ductility

- the ability of a material to undergo large plastic deformation without failure.

$$\%EL = \frac{L_f - L_o}{L_o} \times 100$$



Adapted from Fig. 8.13, *Callister & Rethwisch 9e.*

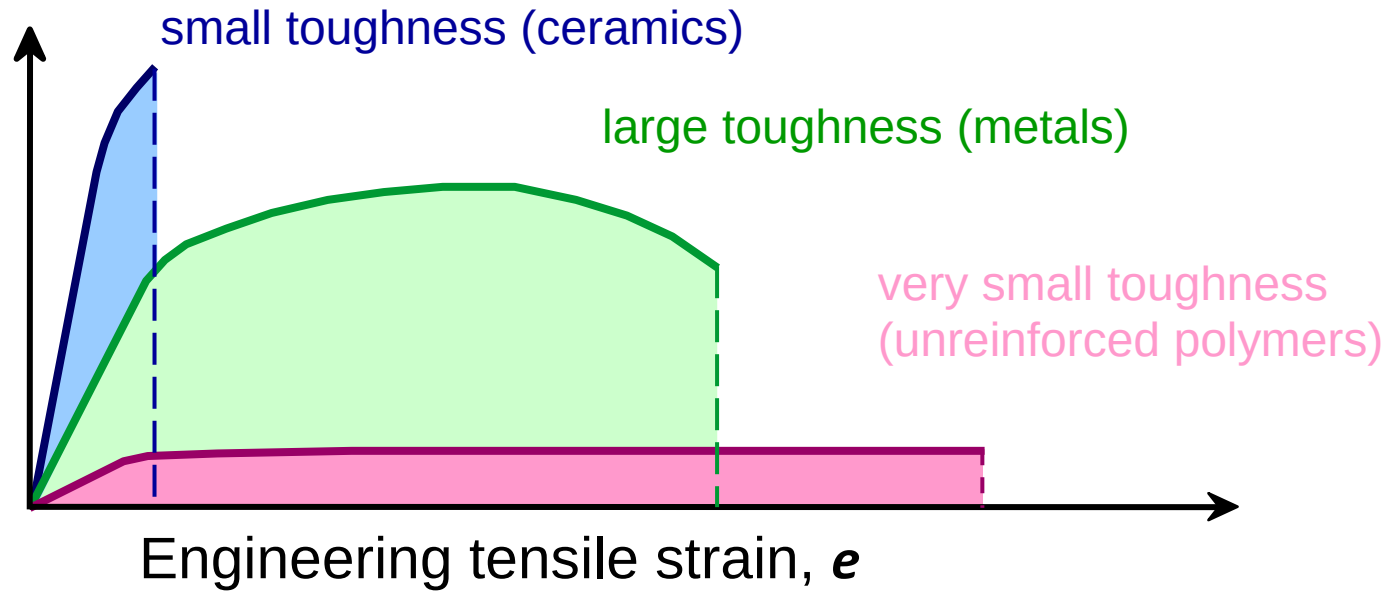
- Another ductility measure:

$$\%RA = \frac{A_o - A_f}{A_o} \times 100$$

Toughness

• the ability to of a material to withstand shock loading without fracture. Energy to break a unit volume of material

• Approximate by the area under the stress-strain curve.



Adapted from Fig. 8.13, *Callister & Rethwisch 9e.*

Brittle fracture: elastic energy
Ductile fracture: elastic + plastic energy

Resilience, U_r

the ability of a material to absorb energy when it is deformed elastically, and release that energy upon unloading

Energy stored best in elastic region

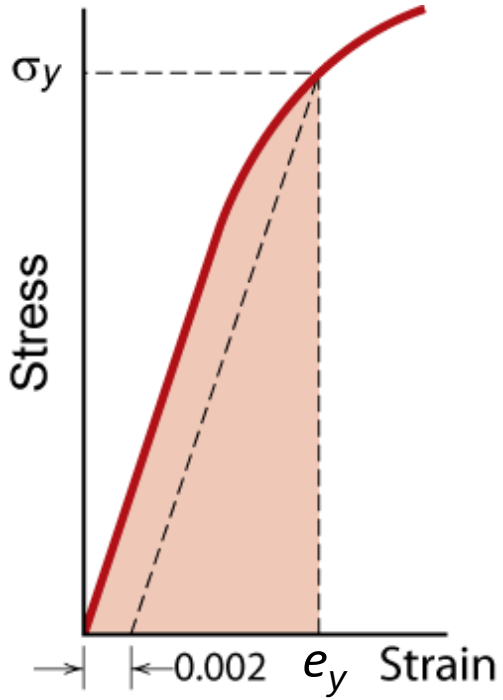


Fig. 8.15, *Callister & Rethwisch 9e.*

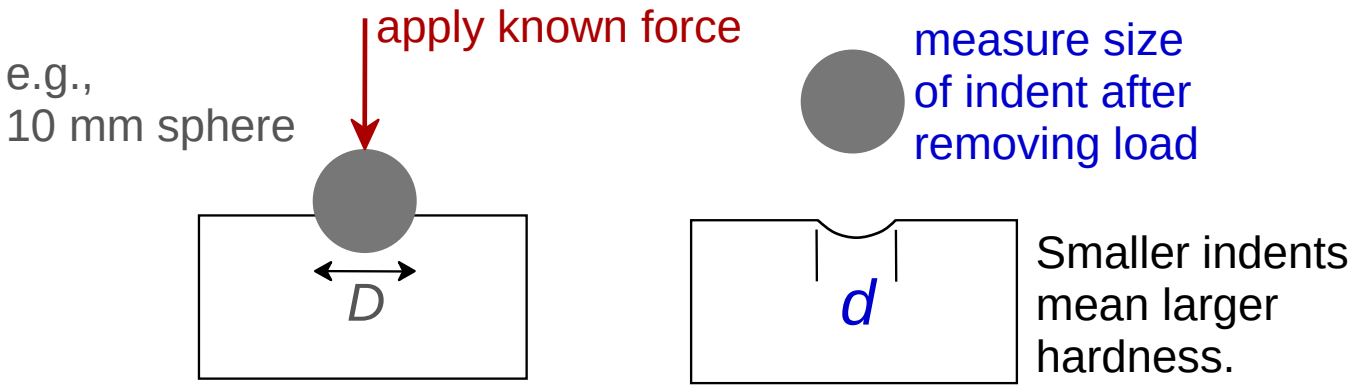
$$U_r = \int_0^{\epsilon_y} \sigma \, d\epsilon$$

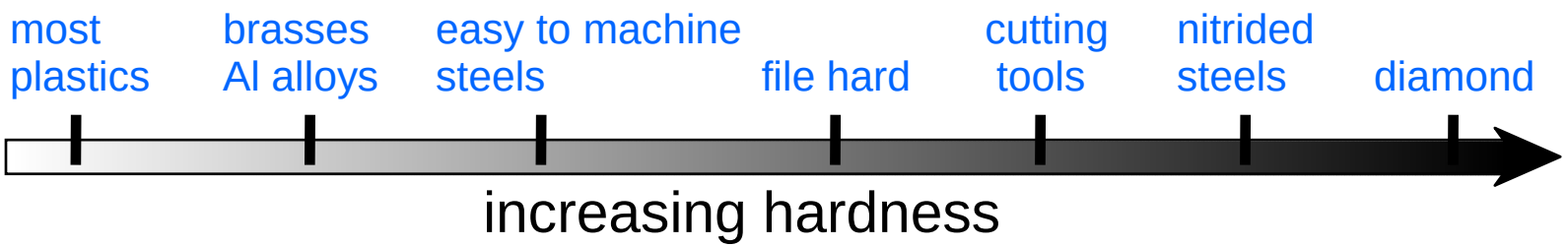
If we assume a linear stress-strain curve this simplifies to

$$U_r \cong \frac{1}{2}\sigma_y e_y$$

Hardness

- Resistance to permanently indenting the surface.
- Large hardness means:
  -- resistance to plastic deformation or cracking in compression.
  -- better wear properties.





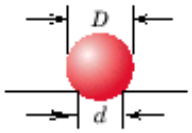
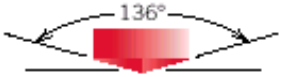
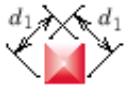
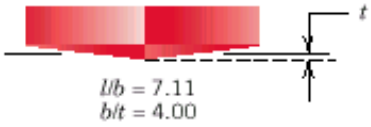
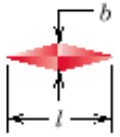
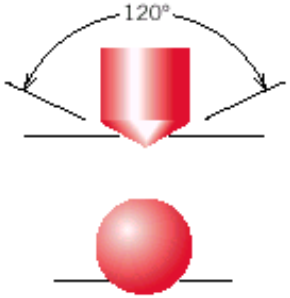
Hardness: Measurement

- Rockwell
  - No major sample damage
  - Each scale runs to 130 but only useful in range 20-100.
  - Minor load 10 kg
  - Major load 60 (A), 100 (B) & 150 (C) kg
    - A = diamond, B = 1/16 in. ball, C = diamond

- HB = Brinell Hardness
  - *TS* (psia) = 500 x HB
  - *TS* (MPa) = 3.45 x HB

Hardness: Measurement

Table 8.5 **Hardness Testing Techniques**

Test	Indenter	Shape of Indentation: Side View	Shape of Indentation: Top View	Load	Formula for Hardness Number[a]
Brinell	10-mm sphere of steel or tungsten carbide			P	$\mathrm{HB} = \dfrac{2P}{\pi D[D - \sqrt{D^2 - d^2}]}$
Vickers microhardness	Diamond pyramid			P	$\mathrm{HV} = 1.854P/d_1^2$
Knoop microhardness	Diamond pyramid			P	$\mathrm{HK} = 14.2P/l^2$
Rockwell and Superficial Rockwell	Diamond cone $\frac{1}{16}$, $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{2}$ in. diameter steel spheres			60 kg, 100 kg, 150 kg: Rockwell 15 kg, 30 kg, 45 kg: Superficial Rockwell	

[a] For the hardness formulas given, P (the applied load) is in kg, while D, d, d_1, and l are all in mm.

Source: Adapted from H. W. Hayden, W. G. Moffatt, and J. Wulff, *The Structure and Properties of Materials*, Vol. III, *Mechanical Behavior*. Copyright © 1965 by John Wiley & Sons, New York. Reprinted by permission of John Wiley & Sons, Inc.

Design or Safety Factors

- Design uncertainties mean we do not push the limit.
- Factor of safety, *N*

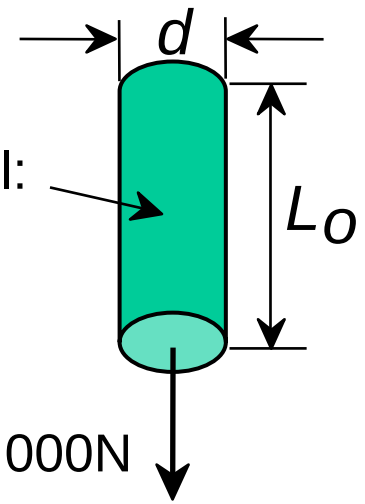
$$\sigma_{working} = \frac{\sigma_y}{N}$$

Often *N* is between 1.2 and 4

- Example: Calculate a diameter, *d*, to ensure that yield does not occur in the 1045 carbon steel rod below. Use a factor of safety of 5.

$$\sigma_{working} = \frac{\sigma_y}{N} \qquad \frac{220{,}000\text{N}}{\pi\left(d^2/4\right)} \qquad 5$$

1045 plain carbon steel:
σ_y = 310 MPa
TS = 565 MPa



d = 0.067 m = 6.7 cm

Summary

- Stress and strain: These are size-independent measures of load and displacement, respectively.
- Elastic behavior: This reversible behavior often shows a linear relation between stress and strain. To minimize deformation, select a material with a large elastic modulus (E or G).
- Plastic behavior: This permanent deformation behavior occurs when the tensile (or compressive) uniaxial stress reaches σ_y.
- Toughness: The energy needed to break a unit volume of material.
- Ductility: The plastic strain at failure.

ANNOUNCEMENTS

Reading:

Core Problems:

Self-help Problems: